



Automation Wiki

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Artifacts

Artifact creation

Automations involved in compiling, packaging, and managing dependencies to generate an artifact.

Code compilation

Maturity level: *Basic*

Implemented by 14% of repositories on GitHub.

Examples that implement this task are:

- Runs "mvn compile"
- Plugin org.apache.maven.plugins:maven-compiler-plugin
- Plugin org.eclipse.tycho:tycho-compiler-plugin
- Plugin org.eclipse.xtend:xtend-maven-plugin
- Runs "./gradlew build"
- Uses gradle/gradle-build-action

Dependency management of artifact

Maturity level: *Basic*

Implemented by 15% of repositories on GitHub.

Examples that implement this task are:

- Plugin org.apache.maven.plugins:maven-dependency-plugin

- Plugin org.eclipse.tycho:tycho-maven-plugin
- Plugin org.codehaus.mojo:versions-maven-plugin
- Runs "npm install"
- Plugin org.apache.maven.plugins:maven-install-plugin
- Plugin org.eclipse.tycho:tycho-p2-plugin

Build tasks, resources and configuration

Maturity level: *Intermediate*

Implemented by 13% of repositories on GitHub.

Examples that implement this task are:

- Plugin org.apache.maven.plugins:maven-resources-plugin
- Plugin org.codehaus.mojo:build-helper-maven-plugin
- Plugin org.eclipse.tycho:target-platform-configuration
- Plugin org.apache.maven.plugins:maven-antrun-plugin
- Plugin org.honton.chas:exists-maven-plugin
- Plugin org.apache.maven.plugins:maven-remote-resources-plugin

Packaging

Maturity level: *Intermediate*

Implemented by 27% of repositories on GitHub.

Examples that implement this task are:

- Plugin org.eclipse.tycho:tycho-packaging-plugin
- Plugin org.apache.maven.plugins:maven-jar-plugin
- Plugin org.springframework.boot:spring-boot-maven-plugin
- Plugin org.eclipse.tycho:tycho-maven-plugin
- Plugin org.apache.maven.plugins:maven-shade-plugin
- Plugin org.apache.maven.plugins:maven-war-plugin
- Runs "./gradlew build"
- Runs "python build"
- Runs "poetry build"
- Runs "mvn package"
- Uses gradle/gradle-build-action
- Runs "zip"
- Runs "tar"

- Uses docker/metadata-action
- Runs "docker build"
- Uses docker/setup-buildx-action
- Uses docker/setup-qemu-action
- Runs "docker run"
- Uses docker/build-push-action
- Runs "docker push"

Release management

Automations related to every aspect of the distribution of artifacts.

Release tagging

Maturity level: *Intermediate*

Implemented by 14% of repositories on GitHub.

Examples that implement this task are:

- Plugin org.apache.maven.plugins:maven-release-plugin
- Plugin org.codehaus.mojo:buildnumber-maven-plugin
- Uses actions/create-release

Publish artifacts to a registry

Maturity level: *Intermediate*

Implemented by 39% of repositories on GitHub.

Examples that implement this task are:

- Runs "mvn deploy"
- Plugin org.apache.maven.plugins:maven-assembly-plugin
- Plugin org.apache.maven.plugins:maven-deploy-plugin
- Plugin org.sonatype.plugins:nexus-staging-maven-plugin
- Plugin org.apache.maven.plugins:maven-release-plugin
- Uses softprops/action-gh-release
- Runs "twine upload"
- Uses actions/upload-artifact
- Uses pypa/gh-action-pypi-publish
- Runs "poetry publish"
- Uses actions/create-release

- Uses docker/build-push-action
- Runs "docker push"

Generate source and metadata artifacts

Maturity level: *Advanced*

Implemented by 7% of repositories on GitHub.

Examples that implement this task are:

- Plugin org.apache.maven.plugins:maven-source-plugin
- Plugin org.eclipse.tycho:tycho-source-plugin
- Plugin org.apache.felix:maven-bundle-plugin
- Plugin biz.aQute.bnd:bnd-maven-plugin

Generate release notes

Maturity level: *Advanced*

Implemented by 1% of repositories on GitHub.

Examples that implement this task are:

- Uses release-drafter/release-drafter

Source control management

Maturity level: *Advanced*

Implemented by 4% of repositories on GitHub.

Examples that implement this task are:

- Plugin io.github.git-commit-id:git-commit-id-maven-plugin
- Plugin pl.project13.maven:git-commit-id-plugin
- Plugin org.apache.maven.plugins:maven-scm-plugin
- Runs "git diff"

Development

Pipeline

Automation workflows used in a runner for execution and optimization of CI/CD pipelines.

Build files configuration

Maturity level: *Basic*

Implemented by 56% of repositories on GitHub.

Examples that implement this task are:

- Uses actions/checkout
- Runs "git checkout"
- Plugin org.apache.maven.plugins:maven-clean-plugin
- Runs "curl"
- Runs "wget"
- Runs "mvn clean"
- Runs "git clone"
- Plugin org.commonjava.maven.plugins:directory-maven-plugin
- Runs "git fetch"

Build environment configuration

Maturity level: *Intermediate*

Implemented by 49% of repositories on GitHub.

Examples that implement this task are:

- Uses docker/setup-buildx-action
- Uses docker/setup-qemu-action
- Uses actions/github-script
- Uses docker/login-action
- Runs "export"
- Runs "source"
- Runs "poetry run"
- Runs "set"
- Uses conda-incubator/setup-miniconda
- Plugin org.codehaus.mojo:flatten-maven-plugin
- Uses actions/setup-python
- Uses actions/setup-java
- Uses actions/setup-node
- Uses docker/setup-qemu-action

Optimization

Maturity level: *Advanced*

Implemented by 14% of repositories on GitHub.

Examples that implement this task are:

- Uses actions/cache
- Uses actions/download-artifact

Code quality

Testing

Run tests

Maturity level: *Basic*

Implemented by 23% of repositories on GitHub.

Examples that implement this task are:

- Plugin org.apache.maven.plugins:maven-surefire-plugin
- Plugin org.eclipse.tycho:tycho-surefire-plugin
- Plugin org.apache.maven.plugins:maven-failsafe-plugin
- Runs "pytest"
- Runs "python pytest"
- Runs "make test"
- Runs "mvn test"
- Runs "./build/mvn"
- Runs "python unittest"

Test coverage and validity

Maturity level: *Intermediate*

Implemented by 9% of repositories on GitHub.

Examples that implement this task are:

- Plugin org.jacoco:jacoco-maven-plugin
- Plugin org.codehaus.mojo:cobertura-maven-plugin
- Runs "coverage run"
- Runs "tox"

Generate test reports

Maturity level: *Intermediate*

Implemented by 10% of repositories on GitHub.

Examples that implement this task are:

- Uses codecov/codecov-action
- Runs "coverage report"
- Runs "coveralls"
- Plugin org.eluder.coveralls:coveralls-maven-plugin

Linting

Static code style analysis

Maturity level: *Basic*

Implemented by 9% of repositories on GitHub.

Examples that implement this task are:

- Plugin com.diffplug.spotless:spotless-maven-plugin
- Plugin org.apache.maven.plugins:maven-checkstyle-plugin
- Runs "flake8"
- Runs "ruff check"
- Runs "pylint"

Automatic code formatting

Maturity level: *Intermediate*

Implemented by 10% of repositories on GitHub.

Examples that implement this task are:

- Plugin net.revelc.code.formatter:formatter-maven-plugin
- Plugin net.revelc.code:impsort-maven-plugin
- Runs "flake8"
- Runs "black"
- Runs "ruff check"
- Uses psf/black
- Plugin com.github.ekryd.sortpom:sortpom-maven-plugin

Static code quality analysis

Maturity level: *Intermediate*

Implemented by 5% of repositories on GitHub.

Examples that implement this task are:

- Plugin com.github.spotbugs:spotbugs-maven-plugin
- Plugin org.apache.maven.plugins:maven-pmd-plugin
- Plugin org.codehaus.mojo:findbugs-maven-plugin
- Plugin org.sonarsource.scanner.maven:sonar-maven-plugin
- Plugin org.codehaus.mojo:animal-sniffer-maven-plugin
- Plugin org.gaul:modernizer-maven-plugin
- Runs "mypy"
- Runs "pylint"

Verify packaging correctness

Maturity level: *Advanced*

Implemented by 6% of repositories on GitHub.

Examples that implement this task are:

- Plugin org.basepom.maven:duplicate-finder-maven-plugin
- Runs "twine check"
- Plugin org.apache.maven.plugins:maven-enforcer-plugin

Security and compliance

Vulnerability scans

Maturity level: *Advanced*

Implemented by 6% of repositories on GitHub.

Examples that implement this task are:

- Uses github/codeql-action/init
- Uses github/codeql-action/analyze
- Uses github/codeql-action/autobuild

Sign artifacts

Maturity level: *Advanced*

Implemented by 3% of repositories on GitHub.

Examples that implement this task are:

- Plugin org.apache.maven.plugins:maven-gpg-plugin

License checks

Maturity level: *Advanced*

Implemented by 2% of repositories on GitHub.

Examples that implement this task are:

- Plugin com.mycila:license-maven-plugin
- Plugin org.apache.rat:apache-rat-plugin
- Plugin org.codehaus.mojo:license-maven-plugin

Collaboration

Documentation

Prepare or create documentation artifacts

Maturity level: *Basic*

Implemented by 18% of repositories on GitHub.

Examples that implement this task are:

- Plugin org.apache.maven.plugins:maven-site-plugin
- Uses actions/upload-pages-artifact
- Runs "make html"
- Uses actions/configure-pages
- Runs "sphinx-build"
- Runs "mkdocs"

Generate documentation from source code

Maturity level: *Intermediate*

Implemented by 7% of repositories on GitHub.

Examples that implement this task are:

- Plugin org.apache.maven.plugins:maven-javadoc-plugin
- Plugin org.apache.maven.plugins:maven-plugin-plugin

Publish documentation

Maturity level: *Advanced*

Implemented by 5% of repositories on GitHub.

Examples that implement this task are:

- Plugin org.apache.maven.plugins:maven-scm-publish-plugin
- Uses actions/deploy-pages
- Uses peaceiris/actions-gh-pages
- Uses Jameslves/github-pages-deploy-action

Social coding

Bot commits

Maturity level: *Intermediate*

Implemented by 6% of repositories on GitHub.

Examples that implement this task are:

- Runs "git commit"
- Runs "git push"
- Runs "git add"
- Runs "git config"

Pre-commit hooks

Maturity level: *Advanced*

Implemented by 4% of repositories on GitHub.

Examples that implement this task are:

- Uses pre-commit/action
- Runs "pre-commit run"

Issue or PR bots

Maturity level: *Advanced*

Implemented by 3% of repositories on GitHub.

Examples that implement this task are:

- Uses actions/stale
- Uses peter-evans/create-pull-request

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